Teacher Wu Kuan | Trig Check — sin(A + B) ≠ sin A + sin B

English lecture notes · YouTube Short b0xEqB_fZC4

1. The trap question

Is sin(A + B) = sin A + sin B?

2. The method

• Test A = 30°, B = 60°: sin 90° = 1, but sin 30° + sin 60° ≈ 1.366.
• Correct: sin(A + B) = sin A cos B + cos A sin B.
• cos(A + B) = cos A cos B − sin A sin B.

3. Parent check method

Have your child plug in 30° and 60° to test any “shortcut” they invent.

Practice

1. sin(30° + 60°) = ?
2. sin 30° + sin 60° = ?
3. Use the sum formula to find sin 75°.
4. Use cos(A + B) with 30° and 60° to find cos 90°.
5. True or false: sin 2A = 2 sin A. Test with A = 30°.

Answers

1. 1
2. (1 + √3)/2 ≈ 1.366
3. (√6 + √2)/4
4. 0
5. False (sin 60° ≈ 0.866, 2 sin 30° = 1)

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